

NOEL EMMANUEL

Senior Backend Engineer

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PROFESSIONAL SUMMARY

Results-driven Senior Backend Engineer with extensive experience designing, building, and maintaining high-scale, highly available API infrastructures. Expert in Node.js, JavaScript/TypeScript, MySQL, and Redis with a strong focus on system performance, speed, and code quality. Proven track record in orchestrating robust backend systems for enterprise platforms and fast-paced food service operations. Adept at resolving complex technical debt, implementing CI/CD pipelines (Jenkins, Docker), and collaborating across product teams to deliver exceptional customer experiences.

CORE TECHNICAL SKILLS

Languages & Frameworks:	Node.js, JavaScript, TypeScript, Express.js, NestJS, Python (FastAPI)
Databases & Caching:	MySQL, PostgreSQL, MongoDB, Redis, Prisma ORM, Mongoose
Cloud & DevOps:	AWS, Google Cloud Platform (GCP), Docker, Jenkins, CI/CD Automation
Architecture & Tools:	REST APIs, WebSockets (Socket.io), Message Queues (Bull MQ), Microservices, JWT Auth, RBAC, Git, Jest

PROFESSIONAL EXPERIENCE & KEY PROJECTS

Lead Backend Engineer – Peregrine POS

Enterprise Point of Sale & Food Service Infrastructure

- Architecture Design:** Built a scalable Node.js/Express backend powering a high-velocity food delivery and restaurant POS system, directly managing real-time order processing.
- Real-Time Order Queues:** Integrated WebSockets (Socket.io) to enable zero-latency, real-time order updates between the customer, vendor, and rider dashboards.
- Performance Optimization:** Designed and implemented a robust Redis caching layer, significantly reducing database load, lowering API latency by over 40%, and ensuring 99.9% uptime during peak food-ordering hours.
- Complex Routing & Security:** Developed strict Role-Based Access Control (RBAC) supporting dynamic navigation and secure endpoints for multiple stakeholders (Admin, Vendor, Rider).
- Production Debugging:** Diagnosed and resolved critical production bottlenecks, reducing app crash reports and ensuring seamless cross-platform integration with mobile endpoints.

Backend Infrastructure Engineer – Ethnikraft Marketplace

High-Scale Three-Sided Marketplace API

- System Design:** Engineered a highly available NestJS backend architecture to power a complex three-sided marketplace, handling intense transactional data and real-time events.
- Asynchronous Processing:** Implemented robust job queues using Redis and Bull MQ to manage heavy asynchronous tasks (emails, notifications, background syncing) without blocking the main event loop.

- **Database Management:** Leveraged Prisma ORM with MySQL/PostgreSQL to execute complex schema designs and queries, optimizing query execution speed by 45%.
- **DevOps & CI/CD:** Spearheaded the automation of deployment pipelines using Jenkins and Docker, saving countless hours of manual DevOps work and eliminating downtime during updates.
- **Quality Assurance:** Maintained a high bar for code quality by implementing comprehensive Unit and E2E testing using Jest, proactively reducing technical debt.

Backend Systems Architect – Enterprise Management System (EMS)

Multi-Location Business Intelligence Platform

- **Scalable Infrastructure:** Built a secure and highly scalable backend supporting multi-location branches, processing large volumes of enterprise analytics and inventory data.
- **Tenant Security:** Engineered secure API endpoints with complex multi-tenant restrictions, ensuring absolute isolation against unauthorized cross-tenant data leaks.
- **Deployment Automation:** Managed Docker containerization and orchestrated automated Jenkins pipelines to streamline staging and production deployments.
- **Data Integrity:** Designed comprehensive database migration scripts and strict rollback safety protocols to prevent data corruption during massive schema updates.